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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Team Bombaklat
Selected Domain	Domain 1 — Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 8 — Decent Work and Economic Growth (proposed primary); SDG 4 — Quality Education (proposed secondary). Confirm the track selection with the team.
Project / Solution Name	SKILLAX AI Platform
Name (Leader)	Cecilia Se
Email & Phone (Leader)	ceciliase1231@gmail.com
University Affiliation(s)	Universiti Sains Malaysia
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Workers, graduates and career changers need to make training and career decisions while occupational skill requirements are changing. The World Economic Forum's *Future of Jobs Report 2025* reports that employers expect 39% of workers' core skills to change by 2030; this is a global employer-survey estimate, not a Malaysia-specific forecast (published 7 January 2025). For Malaysia, Talent Corporation Malaysia's Impact Study examined 949 roles across 10 sectors and estimates that about 620,000 employees, approximately 18% of the workforce in those analysed sectors, may be significantly affected by technological and economic shifts (study launched 18 November 2024). The study also maps pathways for 84 highly impacted roles. These figures describe projected change and study scope; they do not establish that every worker has a skills deficit. SKILLAX addresses a narrower planning problem: an individual may lack a clear, inspectable way to compare their stated skills with a selected role, see the assumptions behind a suggested match, and turn identified gaps into learning steps. The prototype makes those comparisons explicit while distinguishing research activity from job-demand evidence. It has not yet demonstrated improved employment or reskilling outcomes.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SKILLAX is a browser-based career exploration and reskilling prototype for workers, graduates and career changers, with a synthetic-cohort view for workforce planners. Users can enter a profile or try one of three synthetic CV examples; the CV endpoint can accept a PDF or text and, when server-side Gemini credentials are configured, request a structured extraction proposal. An offline keyword extractor is the fallback. Users review the proposal before merging it into their profile. The application compares the profile with ten hard-coded benchmark occupations using transparent TypeScript scoring, shows skill gaps, supports slower, moderate and rapid adoption scenarios, and generates a four-phase learning roadmap with milestone tracking. A dashboard, assessment wizard, progress view, recommendation review history, and SDG mapping are also implemented.

The model is used for CV information extraction only; career scores and rankings are calculated by deterministic functions. The Radar page fetches Hugging Face and MIT RSS items as research-activity signals. Its workforce-demand series are synthetic demonstration data and user-importable observations, not a live Malaysian vacancy feed. User workspace state is stored in browser localStorage; the application has no database. The repository contains a Replit configuration, but a public deployed instance and end-to-end Gemini service call have not been verified for this draft. Phase 2 should test the journey with intended users, replace synthetic demand data only with comparable authoritative data, and evaluate whether roadmap use changes measured skill gaps.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

- **Frontend**: Next.js 14 App Router, React 18, TypeScript, and Tailwind CSS. Versions in `package.json` are ranges; resolved versions are recorded in `package-lock.json`.

- **Server/API**: Next.js route handlers at `POST /api/cv/analyze` and `GET /api/research`.

- **AI**: `@google/genai` SDK. Code selects `process.env.GEMINI_MODEL` or defaults to `gemini-3.5-flash-lite`; this draft verifies the configured identifier and official model documentation, not a successful live API call. Gemini is only used by the CV extraction path.

- **Career logic**: deterministic functions in `lib/domain/` calculate skill gaps, scenario adjustments, match scores, and roadmaps. Baseline: 50% skills, 20% transferable skills, 15% interest, 15% research-activity signal. Where an eligible workforce series is present, overall match combines 90% baseline and 10% workforce score.

- **Data and persistence**: React context and browser localStorage (`skillax_workspace_v2`); no database service or connection is configured. Static catalogs and synthetic workforce observations are bundled in `lib/data/`. Raw CV files/text are posted to the server route; with Gemini configured, the extraction request sends document content to Google's API. Do not describe this as zero external processing or guaranteed deletion by third parties.

- **External data**: live public RSS requests from Hugging Face and MIT AI News for research activity only. Workforce observations are synthetic or user-imported.

- **Hosting**: local Next.js development is documented; `.replit` defines a Replit run/deploy configuration. No public deployment URL or sponsor-provided API/credit is evidenced. Replit use and sponsor-resource status require team confirmation.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The implemented interface provides a profile editor, a five-step assessment, synthetic sample profiles, score explanations, skill-gap views, scenario controls, and a roadmap with user-controlled milestone checkoffs. These choices support exploration without requiring a user to provide a real CV. CV extraction is staged for review rather than automatically merged. The repository contains automated domain tests and project records report 8/8 tests passing; the latest recorded build generated 19 routes. These are software checks, not evidence of user research or usability testing. No user interviews, usability study, pilot, or learning-outcome evaluation is evidenced in the reviewed materials. The prototype runs in a local development setup and includes `.replit` configuration, but public deployment and access by judges are unverified. Next steps are to obtain team and mentor review, test the main journey with a defined participant cohort using synthetic or consented data, deploy to a team-selected platform, and evaluate pre/post skill-gap measures before claiming impact.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Proposed alignment with Domain 1's listed targets: SDG 4.3 (access to technical, vocational and tertiary learning), 4.4 (relevant skills for employment), 8.2 (productivity through technological upgrading and innovation), and 8.5 (productive employment and decent work). SKILLAX supports career exploration and planning but has no measured contribution to these targets. UN target texts: https://sdgs.un.org/goals/goal4 and https://sdgs.un.org/goals/goal8 (accessed 24 October 2026).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Malaysia-focused reference baseline: TalentCorp's study covers 949 roles in 10 sectors and estimates about 620,000 employees (about 18% of the analysed-sector workforce) may be significantly affected. The study launched 18 November 2024. This is a projected exposure estimate, not a measured individual skill gap or a baseline for SKILLAX users. No defensible baseline for the prototype's users' skill gaps is currently available. Source details are cited in Section 2.1. The Domain 1 brief also cites 56% workplace AI-tool use and 26% formal training from Hays Asia Salary Guide 2025; this draft does not rely on these figures until the original Hays source is checked.
Core Measurable KPI *The specific unit you will track*	Mean role-specific skill-gap level per profile, calculated as the sum of positive differences between role-required and user-rated levels across the selected role's listed skills,

Metric Category	Your Response
e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.	divided by the number of listed skills (scale 0–5). Report the cohort mean before and after a defined learning period, plus cohort size and completion rate. Self-ratings are not independent competency validation.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed target for team review: at least 20% reduction in mean role-specific gap level among a defined pilot cohort after a six-week guided-roadmap period, measured using the same skill rubric at baseline and follow-up. Report attrition and use a comparison group if feasible. This is a proposed evaluation target, not a result; the cohort, protocol and percentage require team approval before submission.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React / Next.js	Code Framework	MIT License
Google Gemini GenAI API	Pre-trained AI Model / API	Google Gemini API terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Cecilia Se

24 October 2026